MICHAEL LEHN, Ph.D.

Scientific Software Engineer · C/C++ · Compilers & LLVM · High-Performance Computing

Ulm, Germany · Open to relocation to the United States and Canada

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SUMMARY

Software engineer and mathematician (Ph.D.) with 20+ years of hands-on experience developing performance-critical C/C++ software. Expertise spans numerical algorithms and high-performance linear algebra, memory hierarchy and cache-aware algorithms, SIMD, multithreaded and distributed computing, CPU architecture, compiler construction, and low-level systems programming. Recent projects include an LLVM-based systems programming language with native C interoperability, a minimal self-hosting compiler, and custom CPU architectures implemented on FPGA.

TECHNICAL SKILLS

Languages: C++ (modern, templates/compile-time techniques), C, Python, Assembly (x86-64, AArch64, custom RISC), LLVM IR, SystemVerilog, Java, Perl

Performance & HPC: Numerical linear algebra, BLAS, LAPACK, ScaLAPACK, SIMD micro-kernels (SSE3), cache-aware/blocked algorithms, std::thread, OpenMP, MPI, CUDA, GPU and distributed-memory computing, benchmarking and profiling

Compilers & Systems: LLVM, compiler construction, code generation, x86-64 System V ABI, C interoperability, ISA design, CPU architecture, FPGA (iCE40, ARTIQ/Sinara), bare-metal programming

Tools: Linux/Unix, macOS, Git, Qt, CMake, GNU Make, GDB, Valgrind, perf, Intel VTune, NVIDIA Nsight

SELECTED PROJECTS (OPEN SOURCE)

ulmBLAS **— High-Performance BLAS Library** C++ · Assembly

- Designed and implemented a BLAS library based on the layered architecture pioneered by BLIS, combining portable blocking/packing algorithms in C++ with small, replaceable architecture-specific micro-kernels.
- Reached ~97% of Intel MKL DGEMM performance with BLIS micro-kernels; wrote custom SSE3 micro-kernels and a plug-in interface for externally optimized kernels.
- Drove blocked LAPACK routines (e.g., DGETRF) at performance comparable to vendor LAPACK implementations.
- Authored the GEMM Tutorial on optimizing matrix multiplication from pure C to SSE micro-kernels; referenced by the FLAME project's GEMM optimization resources.
- Used in Meta Quest software; listed in Meta's official third-party software notices.

ABC **("A Better C") — Systems Language & LLVM Compiler** C++ · LLVM

- Designed a small systems language exposing C's machine model with simplified declaration syntax; built an LLVM-based compiler with interchangeable backends targeting LLVM IR and custom ULM assembly.
- Implemented native C interoperability with x86-64 System V ABI lowering, enabling direct use of existing C libraries without wrappers; demonstrated with raylib applications.
- Developed not-abc, a minimal self-hosting compiler originally implemented in ABC, demonstrating the language's suitability for systems programming and compiler construction.

FLENS **— Flexible Library for Efficient Numerical Solutions** C++

- Built a numerical computing framework combining MATLAB-like abstractions with direct access to optimized BLAS/LAPACK kernels, achieving zero abstraction overhead as verified by benchmarks and generated assembly.
- Designed a generic architecture allowing algorithms to operate unchanged on local, distributed, GPU-resident, and implicitly represented matrices.
- Used compile-time C++ techniques and expression analysis to map high-level operations directly to optimized numerical kernels and detect performance-critical temporaries.

ulm-on-iCE / ulm-generator **— FPGA CPU & ISA Toolchain** SystemVerilog · C/C++ · Assembly

- Designed and implemented a custom RISC CPU in SystemVerilog on an iCE40 FPGA, from digital logic and ISA design to assembly programs running on physical hardware.
- Defined a declarative ISA description format and built a generator producing an assembler, virtual machine, instruction-level debugger, and documentation from a single specification.

PROFESSIONAL EXPERIENCE

Ulm University — Institute of Numerical Mathematics · Ulm, Germany 2008 – present

Tenured Faculty Member (Akademischer Rat auf Lebenszeit), Numerical Mathematics

- Develop and maintain open-source numerical and systems software in C/C++: high-performance BLAS/LAPACK kernels (ulmBLAS), a numerical computing framework (FLENS), an LLVM-based systems language and compiler (ABC), and ISA/FPGA CPU tooling (ulm-generator / ulm-on-iCE).

- Own the full engineering lifecycle for these codebases: architecture, implementation, testing, performance benchmarking and profiling, documentation, and long-term maintenance.
- Design and teach courses in high-performance computing, compiler construction, systems programming, numerical mathematics, and scientific computing; the HPC curriculum progresses from memory hierarchy and cache-aware algorithms through SIMD micro-kernels and shared-memory parallelism to MPI and GPU computing.
- Built a full-stack computing curriculum spanning logic gates, registers and ALUs, ISA design, assembly, systems programming, and compiler construction, supported by custom-developed software and FPGA hardware.

Ulm Center for Scientific Computing (UZWR), Ulm University · Ulm, Germany 2006 – 2008
Managing Director

- Developed complete C++/Qt applications for industrial and research partners.
- Translated industrial and interdisciplinary partners' engineering and scientific problems into mathematical models and working numerical software; developed custom solutions where no suitable off-the-shelf software existed.
- Among numerous interdisciplinary projects, designed and developed a self-contained Windows application for MRI-based biomechanical analysis, implementing the complete software stack in-house—from FEM and numerical algorithms to data processing and the graphical user interface.
- Scoped and evaluated incoming projects, designed computational approaches, and either implemented solutions directly or coordinated them with researchers possessing the required expertise.

Ulm University — Institute of Numerical Mathematics · Ulm, Germany 2002 – 2008
Research Associate

- Conducted research and software development in scientific computing and numerical linear algebra, with a focus on high-performance numerical software in C++.
- Designed and developed FLENS (Flexible Library for Efficient Numerical Solutions) as the core of my doctoral research on efficient, abstraction-preserving numerical software in C++.

University of Southern California — Modeling and Simulation Laboratory · Los Angeles, CA 1999 – 2001
Research Assistant / Teaching Assistant

- Conducted research in applied mathematics and scientific computing with Prof. Chunming Wang in USC's Modeling and Simulation Laboratory while completing an M.S. in Applied Mathematics; master's research conducted in collaboration with NASA's Jet Propulsion Laboratory (JPL).

SELECTED INDUSTRY & RESEARCH COLLABORATIONS

German Aerospace Center (DLR) — Real-Time Control / Quantum Technologies 2025 – present

- Collaborate with the DLR Institute of Quantum Technologies on real-time control for a space-ready quantum-memory system based on atomic-gas technology, using ARTIQ/Sinara with an FPGA-based core device.
- Develop HeraldRT, an FPGA-based real-time controller in SystemVerilog that maps the physicists' timing requirements to deterministic hardware. Simulated, synthesized, placed, and routed; hardware validation pending.
- Contribute to the architecture of the real-time control system and scientist-facing software: state-machine control, experimental parameter management, measurement and imaging acquisition, and hardware integration.

Tensor AI Solutions — HPC / Tensor Networks 2024

- Collaborated with Tensor AI Solutions on the design and development of Tartarus, a C++ HPC library for efficient tensor contractions in tree-tensor-network applications for machine learning and QML.
- Co-supervised a research-driven student project, contributing substantially to the numerical algorithms, performance-oriented design, and development of the library.
- Applied TTGT and Loops-over-GEMM (LoG) contraction schemes, BLAS kernels (GEMM, GEMV, GER, DOT), memory-layout-aware algorithms, and Python-based C++ code generation to optimize tensor contractions on CPUs.

Daimler TSS (Daimler Group) — Machine Learning / Recurrent Neural Networks 2017 – 2018

- Collaborated closely with Daimler TSS on the mathematical analysis of recurrent neural networks, with a particular focus on training and the vanishing/exploding gradient problem.
- Co-supervised a research-driven bachelor's thesis, contributing substantially to the mathematical formulation, analysis, and development of the work.
- Applied regression and optimization methods, gradient-based training and backpropagation, and mathematical analysis of feed-forward and Elman recurrent neural networks.

EDUCATION

Ph.D., Numerical Mathematics — Ulm University, Germany 2008
Dissertation: FLENS — A Flexible Library for Efficient Numerical Solutions · Advisor: Prof. Karsten Urban

M.S., Mathematics — Ulm University, Germany 2002

M.S., Applied Mathematics — University of Southern California, Los Angeles, CA 2001